

HK IMO Training 2020-2021
Problem Set 15
For 20 February 2021



Problem 57. Let $ABCD$ be a convex quadrilateral such that the sides BC and CD have equal length and $2\angle A + \angle C = 180^\circ$. Let M be the midpoint of BD . Prove that $\angle MAD = \angle BAC$.

Problem 58. Let ABC be a triangle and BE, CF be the altitudes from B, C respectively. Let Y and Z be the midpoints of CE and BF respectively. The point P is an arbitrary point on the perpendicular bisector of BC . Let Q be the intersection of the line through B perpendicular to PZ and the line through C perpendicular to PY . Prove that AQ is the symmedian of $\triangle ABC$.

Problem 59. Let $ABCD$ be a quadrilateral inscribed in a circle ω . The symmedians through B of $\triangle ABD$ and $\triangle CBD$ intersect ω again at points P and Q , respectively. Lines CP and AB intersect at X , and lines AQ and BC intersect at Y . Prove that the points X, D, Y are collinear.

Problem 60. Let ABC be a fixed acute-angled triangle. Consider some points E and F lying on the sides AC and AB , respectively, and let M be the midpoint of EF . Let the perpendicular bisector of EF intersect the line BC at K , and let the perpendicular bisector of MK intersect the lines AC and AB at S and T , respectively. We call the pair (E, F) *interesting*, if the quadrilateral $KSAT$ is cyclic.

Suppose that the pairs (E_1, F_1) and (E_2, F_2) are interesting. Prove that $\frac{E_1E_2}{AB} = \frac{F_1F_2}{AC}$.